

Hostel = State 1 LHC = State 2

Library = State 3

$$V_i = \max_a \sum_{s'} P(s'|s, a) [R(s, a) + \gamma V(s')]$$

$$Q(s, a) = R(s, a) + \gamma \max_{a'} V(s', a')$$

Let travel(hostel) = a_1 , travel(LHC) = a_2

Stay(LHC) = a_3 , Decide & act(Library) = a_4

Iteration

①

Initially $V_0(1) = 1$ $V_0(2) = 0$ $V_0(3) = 0$

~~Starting at Hostel, looking at LHC~~

$$V_1(1) = 0$$

↖

$$V_1(2) = \text{stay} \Rightarrow 0.4 \times 5 + 0.3 \times 0$$

$$V_1(2) = 1.7$$

Iteration

②

$$\text{Library} \Rightarrow V_1(3) = 0.7 \times 5 = 3.5$$

$$V_2(1) = [0.8 \times 1.7 + 0.2 \times (3.5)] \times 0.9 = 2.06 \times 0.9 = 1.854$$

$$V_2(2) = \max \{ 0.7 \times 3.5 \times 0.9, 1.7 + 0.9 \times 1.7 \} = \max \{ 2.2, 3.23 \} = 3.23$$

Iteration

③

$$V_2(3) = 0.7 \times 5 + 0.1 \times 0.9 \times 0 = 3.5$$

$$V_3(1) = [0.8 \times 3.23 + 0.2 \times 3.5] \times 0.9 = 2.95$$

$$V_3(2) = \max \left\{ \underset{3.5}{0.7 \times 3.5} + 0.3 \times 1.85 \right\} \times 0.9, 1.7 + 0.9 \times (3.23) \} = \max \{ 2.7, 4.6 \} = 4.6$$

$$V_3(2) = 4.6$$

$$V_3(3) = 0.7 \times 5 + 0.1 \times 0.9 \times (1.85) = 3.66$$